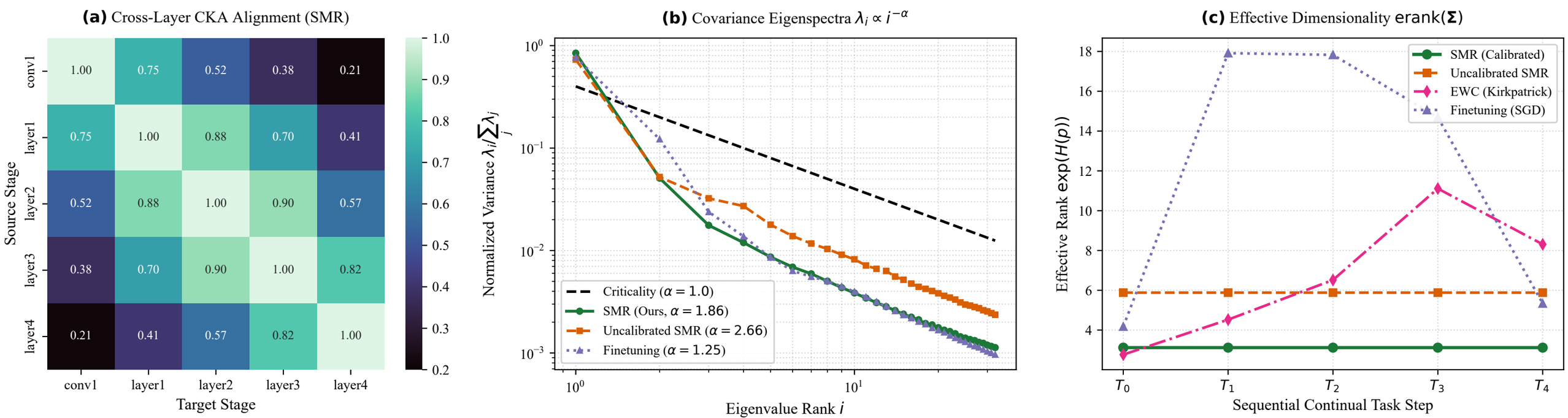

(a) Cross-Layer CKA Alignment (SMR)
Source Stage
Target Stage
conv1
layer1
layer2
layer3
layer4
1.00 0.75 0.52 0.38 0.21
0.75 1.00 0.88 0.70 0.41
0.52 0.88 1.00 0.90 0.57
0.38 0.70 0.90 1.00 0.82
0.21 0.41 0.57 0.82 1.00
1.0 0.9 0.8 0.7 0.6 0.5 0.4 0.3 0.2
(b) Covariance Eigenspectra $\lambda_i \propto i^{-\alpha}$
Normalized Variance $\lambda_i / \sum_j \lambda_j$
Eigenvalue Rank i
Criticality ($\alpha = 1.0$)
SMR (Ours, $\alpha = 1.86$)
Uncalibrated SMR ($\alpha = 2.66$)
Finetuning ($\alpha = 1.25$)
(c) Effective Dimensionality erank(Σ)
Effective Rank $\exp(H(p))$
Sequential Continual Task Step
SMR (Calibrated)
Uncalibrated SMR
EWC (Kirkpatrick)
Finetuning (SGD)
T_0 T_1 T_2 T_3 T_4